

Fractions: Division

Dividing Fractions

Definition 1 (Reciprocal of a Fraction). The **reciprocal** of the fraction $\frac{a}{b}$ is $\frac{b}{a}$.

Remark 1. (a) The reciprocal of a fraction is the fraction with the numerator and denominator interchanged (inverted).

(b) The **reciprocal of an integer** is found by writing the integer over 1 and then interchanging the numerator and denominator. For example, to find the reciprocal of -4 , write it as $-\frac{4}{1}$, then interchange the numerator and denominator to get $-\frac{1}{4}$.

(c) Two numbers are reciprocals of each other if their product is 1.

$$\frac{a}{b} \cdot \frac{b}{a} = 1$$

(d) Every number has a reciprocal except 0. This is because if we write 0 as $\frac{0}{1}$ and then take the reciprocal we get $\frac{1}{0}$ which is undefined. Furthermore, if 0 had a reciprocal then the product of 0 and its reciprocal should equal 1, but we know that the product of 0 and any number equals 0.

Problem 1 Find the reciprocal of $\frac{6}{11}$.

$$\frac{11}{6}$$

Problem 2 Find the reciprocal of 18.

$$\frac{1}{18}$$

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Definition 2 (Dividing Fractions). If a, b, c and d are numbers and b, c and d are not 0, then

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \cdot \frac{d}{c}$$

To divide with fractions, multiply the first fraction by the reciprocal of the second fraction.

Warning 1. DO NOT SIMPLIFY until you have written the division problem as an equivalent multiplication problem. In the division problem below, you might think that the fives cancel, however by rewriting it as multiplication, you can see that they do not.

$$\frac{2}{5} \div \frac{5}{3} = \frac{2}{5} \cdot \frac{3}{5}$$

Problem 3 Divide. Remember to simplify your answer, if possible.

$$\frac{10}{11} \div \frac{4}{5} = \frac{25}{22}$$

$$7 \div \frac{42}{3} = \frac{1}{2}$$

$$\frac{9}{8} \div \frac{3}{7} = \frac{21}{8}$$

Problem 4 You have a ribbon that is $\frac{5}{6}$ of a yard long. You need to cut the ribbon into shorter pieces, each measuring exactly $\frac{1}{12}$ of a yard. How many pieces can you cut from the ribbon? Begin by setting up the division problem, then find the answer.

Multiple Choice:

(a) $\frac{5}{6} \div 12$

(b) $\frac{5}{6} \div \frac{1}{12}$ ✓

(c) $\frac{1}{12} \div \frac{5}{6}$

(d) $12 \div \frac{5}{6}$

10 pieces

Hint: Divide the total length by the length of each piece.